



ARTIFICIAL INTELLIGENCE STRATEGY FOR MOHRE

Developing a Roadmap for Leveraging AI Benefits

ABSTRACT

The successful operational use of artificial intelligence (AI) today depends on understanding the value, objectives and inputs needed for the decision that drives the use of AI. Accomplishing a successful AI project requires skill, resources and knowledge of AI tools and methods.

Knowledge Consultants, Inc.



Executive Summary

Organizations today have the flexibility to deploy artificial intelligence (AI) incorporating capabilities such as decision analysis (DA), deep learning (DL) machine learning (ML) and knowledge based (KM) and expert systems (ES) at well controlled (and managed) points throughout their organization.

With such a variety of available capabilities, bringing AI into MOHRE requires careful thought and a good plan for introducing the technology into MOHRE. An organization unfamiliar with the tools, methods and algorithms of AI needs a staged approach to avoid the hazards of embarking on a direction that is less than satisfactory to performance and in some cases very costly with little benefit.

A gradual and evolutionary approach is suggested that uses an introductory stage and a several expansion stages. In most cases that requires at least a 2 to 3 stage start up used to establish a path for forward planning. A multi stage approach starting with 3 stages might contain the following:

Stage 1 – The short plan: The focus is on learning and knowledge transfer that identifies the value points for AI and highlights some starter projects to assess the value of applying AI types of analytics to the organization. Stage one is used when knowledge is limited, and the execution time frame is short like for part of a year.

Stage 2 – The implementation for proof of concept: This stage includes implementation and evaluation of pilot projects, usually 3 to 6 small projects, to establish the proof of value as well as proof of concept for MOHRE.

Stage 3 –The longer view forward plan: Based on the results of the first 2 stages, a longer view 5 to 20 years is developed as a 'living' plan that gets updated each year based on experience, organization need, results of Stage 1 and changes on the organization landscape.

Long term efforts are designed to assure the sustainability and growth of an organization in the future. For the AI effort, direction, value and risk show up during and through Stage 1 and 2 and continue and accelerate during Stage 3. Annual or even semi-annual review assures that the path is meaningful and viable and meets the needs of the organization. Long range forward plans are done during the normal planning cycle of an organization

Project Effort:

Stage 1 – Short term plan that identifies possible quick hits that are easy to implement and provide value. Determination is done by using risk/opportunity analytics and ranking analytics to position opportunities according to what is known at the time of the analysis. Analysis is limited to the high hits keeping the budget reasonable during the learning process. Stage 1 ends with an evaluation of the results. An initial suite of 3 to 6 ‘quick hit’ projects are identified along with the resources needed for implementation.

Stage 2 – Using as input to Stage 2 the evaluation of Stage 1 two paths are established:

- A) Implementation of the initial set of AI applications following the basic steps of AI implementation described below.
- B) A forward plan is developed that identifies a second suite of AI applications. This plan includes the assets that are needed to maintain an ongoing AI effort that aligns with country goals for AI and produces expected value to the organization. This stage includes identification and planning for skills, knowledge, infrastructure and focus of the AI effort for MOHRE

Stage 3 – This stage involves the roll out of multiple AI projects across the operational processes of HR. The focus of AI effort here is on machine learning, decision analysis, knowledge and expert systems and horizontal integration of AI capabilities as appropriate. Big data and deep learning are marginally applicable at this level and are applied only where Commercial Off The Shelf (COTS) solutions are available.

Stage 4 and beyond – Vertical AI integration is accomplished through the use of measure and performance metric integration using AI capabilities especially neural nets for predictive analytics and deep learning.

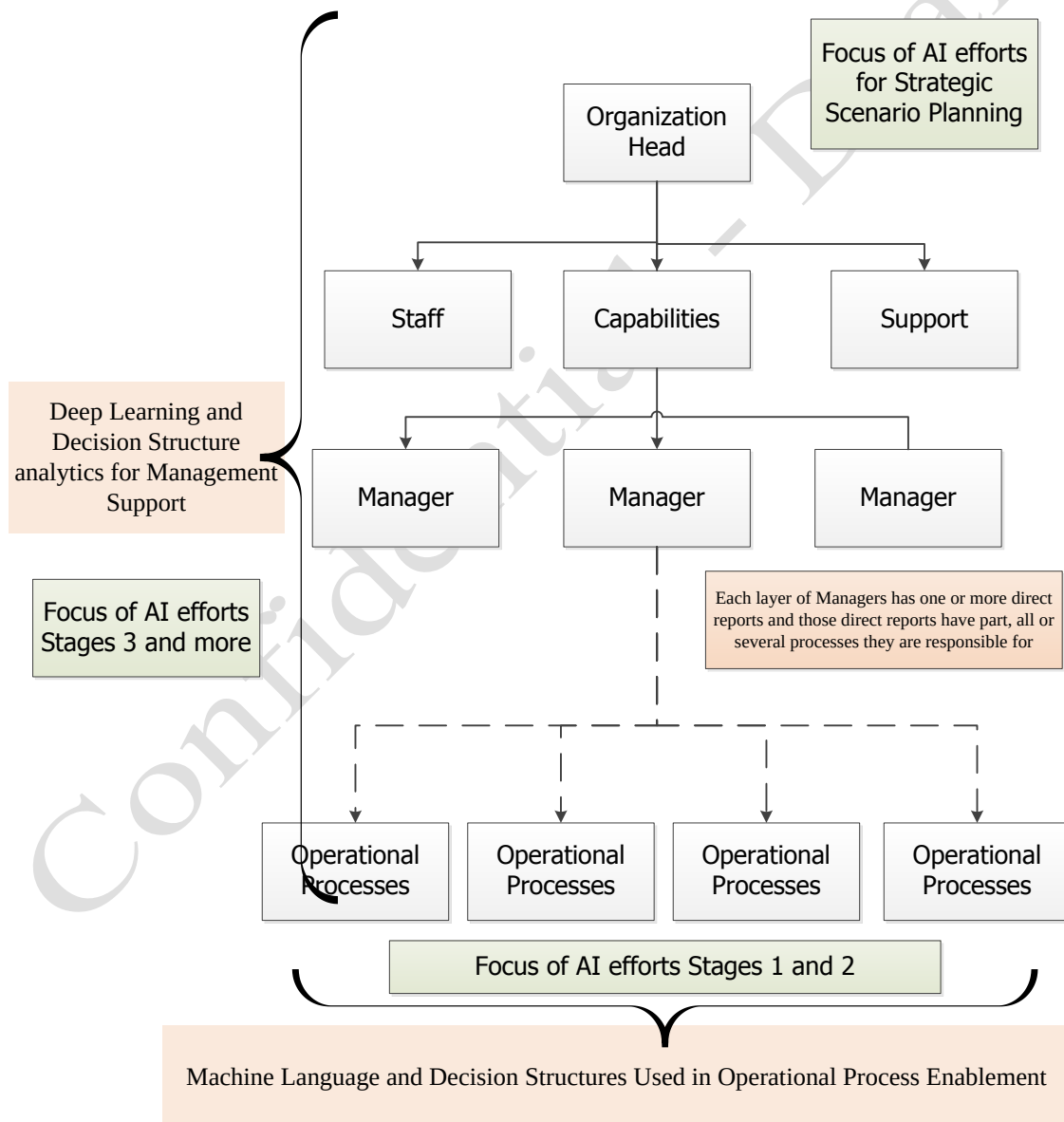
A key part of this project is *knowledge transfer*. A client team (4 to 5 people) is critical to the success. The team mix should include at least one person from IT, one management representative and 2 or 3 functional people with process knowledge. Another key part is the need to start small and safe, so the staff can learn the key elements of using AI.

The focus of the initial MOHRE effort is on stages 1 and 2.

The Long Term Vision of AI Transformation

There is a core methodology (proprietary to Knowledge Consultants, Inc.) that supports an evolutionary introduction of AI into an organization. The diagram below highlights the key points of the methodology. The method is divided into major stages and reflects how an AI mature organization would manage and AI project.

Initial AI Application Perspective: Focus on Machine Language and Decision Structures



The perspective of evolutionary AI described here takes a number of years for fully realized results. It starts small with some simple applications to get value from the initial efforts, builds capability and evaluates and selects proper tools that support the direction an organization wants to pursue. Transformation grows first horizontally and then vertically through the process and management structure. The goal is to start simple and build on the tools, skills and expertise the results from the startup.

Several of the concepts here come from the KCI history of the past 30 years starting in process automation, supervisory control, numerical control of tools (early AI), robotic control systems for manufacturing, expert systems projects, text analysis and analytics, predictive neural nets (mainly in oil and gas industry), business intelligence and business analytics systems for senior management. Large scale long term projects (5 to 25 years) require careful consideration in their planning and execution.

There are 3 core focal points of evolutionary AI change for an organization:

1. Enablement of operational processes (Stages 1 and 2):

Initial opportunities are driven by identified needs in the basic operational processes of an organization. The majority of the applications are based on machine learning. The machine learning algorithms typical of supporting processes include statistical techniques used for prediction such as:

- Correlation and regression,
- Correlation matrices,
- Affinity analysis,
- Sentiment analysis,
- Keyword analysis
- Clustering
- Spine analysis (such as used for population graphics)
- Influence diagrams
- Decision trees
- Monte Carlo simulation

In certain cases there may also be deep learning AI applications where the data input is large and a large degree of classification and sorting are required. In the HR environment this might include sorting through hundreds of applicants, millions of citizens such as relating to Emiratization, sorting through resumes and many other tasks that require narrowing down the inputs based on a set of criteria established by HR.

A set of easily available tools are used in this effort and the market has a number of commercially available deep learning application focused tools like sorting through resumes or standard chatbots for contacting applicants for key HR applications.

There are commercial HR applications for text recognition, speech recognition, vision recognition, chatbots and other data intensive applications that are on the market today. Many of them work quite well but all have some issues. Also, they may demand considerable infrastructure or require use of cloud based resources.

The increasing use of AI process augmentation requires considerations of horizontal integration of AI capabilities and outputs. Tools are available to support this effort also. Later stages of the AI transformation will require such tools.

2. Leveraging decision making of managers (Stage 3 and beyond):

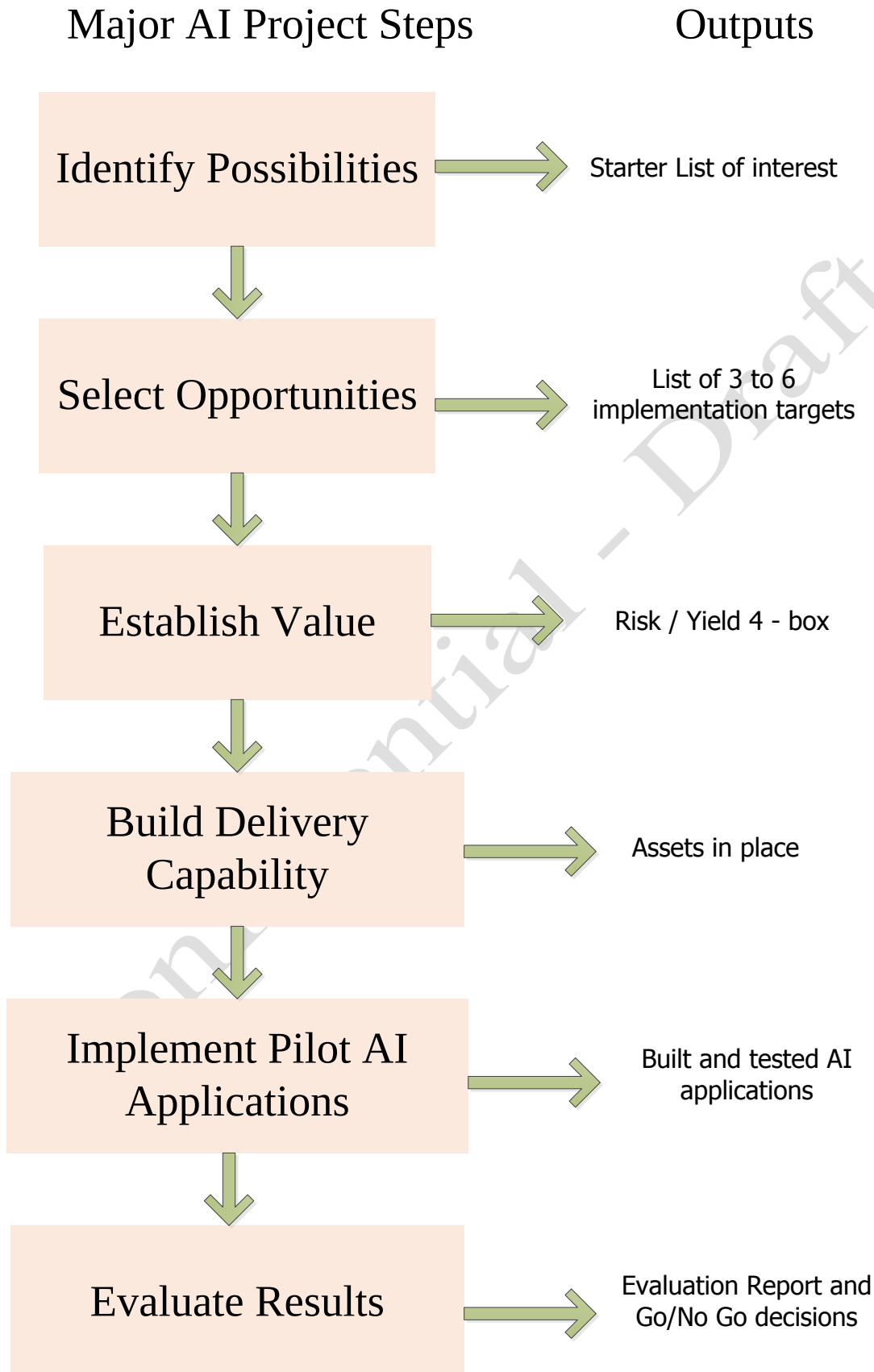
With the automation of more and more operational components the role of both professionals and managers will change. The need for horizontal integration across processes and AI applications in a manner suitable for a manager to make decisions about operations emerges. Fortunately, there are tools emerging in this domain also that can support the management of this new environment.

As the first layer or two of managers gets more involved it becomes necessary to look at integrating the AI transformation vertically. Automated feeds into tools that are called 'Management Cockpits' provide the manager with a view into how performance is evolving in their scope of reference. These tools are part of a new generation that is emerging now. They include tools for streaming analytics (e.g. Data Torrent), digital business capability (e.g. IYCON – Afterspyre), robotic process automation (e.g. Blue Prism), talent management (e.g. Intuo), learning (e.g. Area9), advanced chatbots (e.g. Azumo) and content management (e.g. NUXEO).

Stage 3 and beyond means that the effort will extend over several years as each layer of management identifies their needs and learns how to take advantage of the AI capability.

3. Supporting senior management planning (Vertical integration of AI management augmentation):

The planning and overall management advantage comes from integrating the senior management view into the operations of the organization. The prospect of an AI linkage gives the management a more 'hands on' approach to general direction of the organization.



AI/Machine learning and Deep learning projects share a basic set of steps. These steps are *executed for each AI project* regardless of size. There are 10 basic steps to an AI project journey, namely:

1. Objective identification – the organization opportunity such as augmenting recruitment, onboarding, evaluation, selection and other stages of employee acquisition.
2. Input identification (parameters that drive to the objective) often using a driver diagram
3. Data discovery, sourcing and gathering
4. Data preparation
5. Choosing the model (type may vary from project to project, hence there may be more than one type of tool, Neural net, decision structures, business analytics etc.)
6. Training the net or testing the statistical machine learning model
7. Evaluating the results
8. Tuning the parameters (Increasing the number, decreasing, removing assessing impact)
9. Prediction via some form of inference from inputs to outputs
10. Evaluation of value and impact

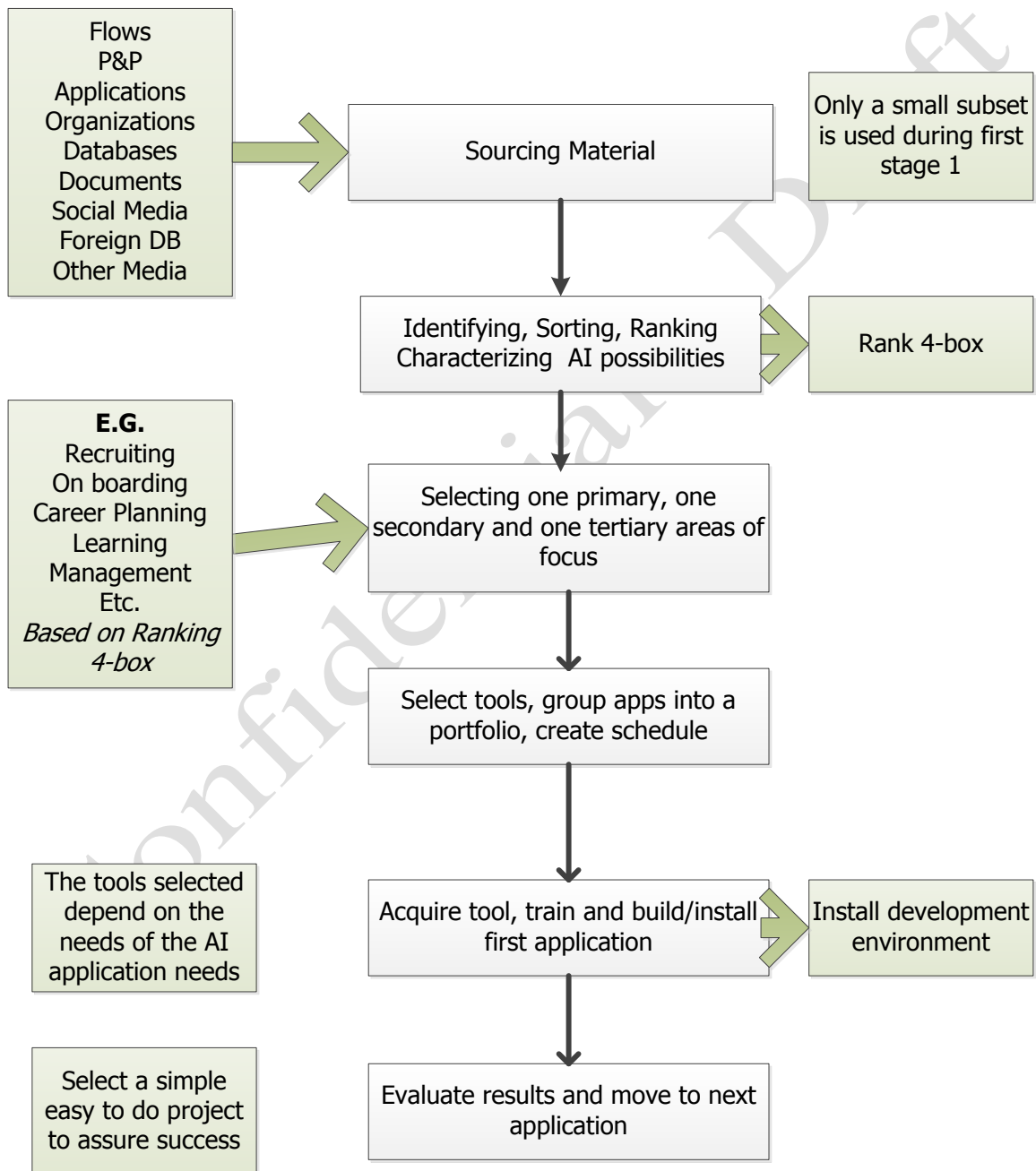
Each project selected for implementation will go through these 10 steps. Some projects will take more time than others in certain steps depending on such factors as amount of data, availability, type of algorithms, tools available and so on.

What do you get from applying AI to process augmentation? Here are some result types that organizations have reported:

- Faster decision making, especially for sorting through applicants and resumes, scheduling, and evaluation.
- Lower operating cost or increased throughput for the same cost. Some of the cost advantage comes from reduced staff. Most organization re-assign staff to more productive positions.
- Improved quality of results, lower error rates
- Increased positive sentiment from customers or citizens
- Reduced waste especially in topics like benefits management
- Improved insight into labor issues.

This is a basic overview of the initial project efforts. The goal is to use a subset of available source material based on meetings with the AI project team and management. The selection of implementation targets focuses on value, usefulness and ease of implementation. That said, they will probably use acquired application products so the focus is on data prep and outputs needed.

Core AI Project Steps



Project schedules:

Two schedules are required to manage a program that has a suite of projects:

1. Stages of overall project progress
2. Implementation of the individual AI projects

Stages of overall project progress

A schedule like the one shown below is prepared for the first project. After that a modified version is prepared for each of the 3 to 6 projects selected. These form the basis of the AI portfolio.

ID	Task Name	Start	Finish	Duration	Jun 2018				Jul 2018				Aug 2018				Sep 2018				Oct 2018				Nov 2018				Dec 2018			
					6/3	6/10	6/17	6/24	7/1	7/8	7/15	7/22	7/29	8/5	8/12	8/19	8/26	9/2	9/9	9/16	9/23	9/30	10/7	10/14	10/21	10/28	11/4	11/11	11/18	11/25	12/2	12/9
1	Sourcing material	6/1/2018	7/13/2018	31d																												
2	Rank Opportunities	7/2/2018	7/30/2018	21d																												
3	Select focus area for first projects	7/2/2018	7/30/2018	21d																												
4	Identify 3 to 6 target projects	8/1/2018	8/15/2018	11d																												
5	Identify related COTS tools	8/1/2018	8/30/2018	22d																												
6	Create risk/yield 4-box	8/15/2018	8/30/2018	12d																												
7	Prepare implementation plan	8/15/2018	8/30/2018	12d																												
8	Acquire tools	8/15/2018	9/14/2018	23d																												
9	Implement project	9/3/2018	10/30/2018	42d																												
10	Install, Test and evaluate	10/30/2018	12/28/2018	44d																												

Implementation progress of a suite of AI projects

Implementing the initial suite of pilot projects requires basic project management including a simple project plan. A classic Gantt chart is used to manage the suite of project. A typical AI application development portfolio has a chart that looks like this:

ID	Task Name	Start	Finish	Duration	2018																													
					Jun 2018				Jul 2018				Aug 2018				Sep 2018				Oct 2018				Nov 2018				Dec 2018					
					6/3	6/10	6/17	6/24	7/1	7/8	7/15	7/22	7/29	8/5	8/12	8/19	8/26	9/2	9/9	9/16	9/23	9/30	10/7	10/14	10/21	10/28	11/4	11/11	11/18	11/25	12/2	12/9	12/16	12/23
1	Filter recruits	6/1/2018	6/29/2018	21d																														
2	Create target applicant list	7/2/2018	8/31/2018	45d																														
3	Improve on-boarding	6/15/2018	8/15/2018	44d																														
4	Improve Security	6/15/2018	9/14/2018	66d																														
5	Improve Learning Results	7/2/2018	10/2/2018	67d																														
6	Identify employee needs	8/16/2018	11/16/2018	67d																														
7	Increase employee retention	9/17/2018	12/17/2018	66d																														

Risk Assessment for AI Projects

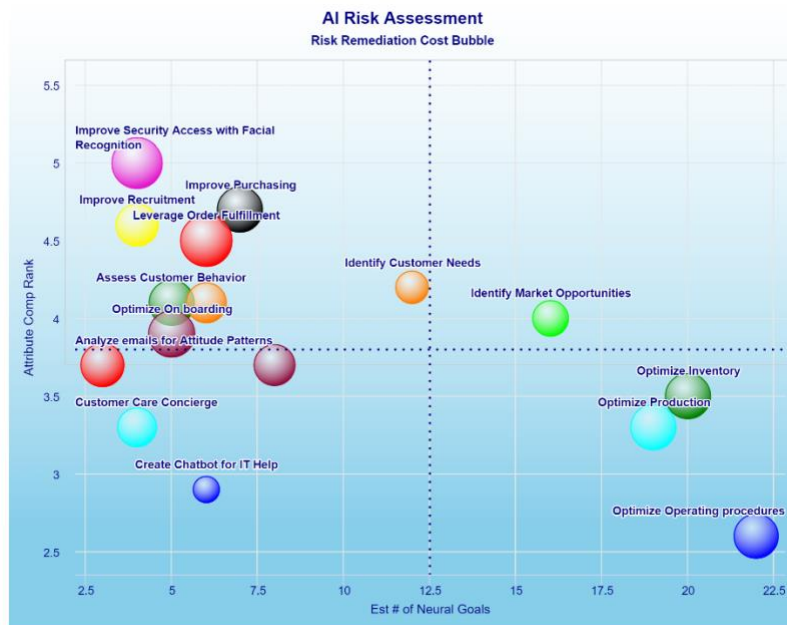
Understanding the risk – benefit of a suite of projects is also key to ranking the opportunities. A set of application need and desirability attributes are used to create a composite ranking of the significance and importance of the each of the AI projects. This attribute list includes but is not limited to the following:

- Benefit
- Impact
- Importance

Along with the identified desirable AI projects attributes, a list of risk attributes is developed and used to produce a risk ranking. A typical set of attributes might include but not be limited to the following:

- Number of objectives
- Availability of data
- Quality of the data
- Skills to for the AI method
- Time
- Funding

The paired ranking are used to create a four box assessment regression to identify the ‘quick hits’ and rank projects. An example of such a four box for AI projects is shown below.



Applying AI to an HR organization:

The typical HR organization consists of a number of functions, services that are supplied to the parent organization. Among these usually considered for AI augmentation and used as discovery of points of AI opportunity are the following functions:

- Employee Lifecycle Management
 - Onboarding
 - Location Change
 - Role Change
 - Name Change
 - Termination
- Recruiting Management
 - Sourcing (Direct & Indirect)
 - Acquisition Management
 - Interview / Selection Management
 - Vendor Management (3rd party)
 - Talent Pool Management
- Career Management
 - Career Planning
 - Career Development
- Learning Management
 - Skills enhancement
 - Rotation Management
 - OTJ / Apprenticeship
 - Mentoring & Guidance Programs
 - Executive Coaching
 - Leadership Development
 - Compliance Training
 - Certification Management
- Organization Management

- Organization Design (model)
- Organization Transformation
- M&A Planning
- Outsourcing (offshore)
- Outsourcing (machines)
- Benefits Management
 - Healthcare Insurance Management
 - Healthcare Plan Acquisition
 - Healthcare Plan Monitoring
 - Healthcare Plan Management
 - Healthcare Plan Performance
- Benefits Management (cont.)
 - Life & Disability Management
 - Life Insurance Plan (A, M, M, P)
 - Disability Insurance Plan (A, M, M, P)
 - Maternity Leave Management
 - Out of Office Management
 - Planned (vacation)
 - Unplanned
 - Holiday Planning Management

Additionally, there are a number of cross functional support tasks needed to make the HR organization work efficiently and effectively, namely:

- Data Management
- Customer Management
- Project Management
- Workflow / Event Management
- Content / Document Management
- Employee / Vendor Management
- Decision Management (analysis, reporting, metrics)
- Compliance Management
- Financial Management

The results of the assessment is a table of what AI features and capabilities might best be applied to a particular portion of the organization based initially in process enablement. Below is an example of one way to identify and characterize the AI need. In the actual project specific algorithms would be identified that relate to the project. This simple matrix just indicates which features of an AI implementation a particular part of the organization might need.

D Level	Domain Name	Service	AI, ML, DL	RPA/RSA	Micro – Services	A/S
2	Organization Management		Y, DL			Y
3		Organization Design (model)	Y, ML			Y
3		Organization Transformation	Y, ML	Y	Y	Y
3		M&A Planning	Y, ML			Y
3		Outsourcing (offshore)		Y		
3		Outsourcing (machines)	Y, ML	Y	Y	

RPA/RSA – Robotic Process Automation/Robotic Service Automation

AI – Artificial Intelligence

ML – Machine Learning

DL – Deep Learning

A/S – Afterspyre Product Suite for Management

Detailed Approach

- a. *Stage 1* – Development of short term plan, we would do the following steps with the AI team:
- b. Conduct a short training session on AI and the planning for AI transition for MOHRE (2 days, 10 to 12 people, the team is selected from these individuals)
- c. Obtain any material from UAE AI Ministry that may impact the AI efforts of MOHRE

Alignment of MOHRE strategy with the government's AI Ministry.

- d. Study of current processes and current data to source AI material
 - i. Processes and MOHRE functions as potential targets
 - ii. Identification of objectives (one or more) for each identified target
 - iii. Identification of inputs (factors) to solve the target
 - iv. Identification of primary and secondary AI strategy for each target, e.g.
 1. Neural nets
 - a. Predictive
 - b. Classification
 2. Decision and influence structures
 3. Statistical machine learning algorithms (there are about 100 of these to choose from) but the key ones here are:
 - a. Regression following prediction and/or forecasting
 - b. Clustering
 - c. Dimensionality
 4. Others as needed
 - v. Identification of attributes of each AI project
- e. Construct the models and apply the analytics that sort out the opportunities and identify the short-term hits.
 - i. Preparation of starter models for evaluating and ranking opportunities based on:
 1. Risk
 2. Yield

3. Importance
4. Impact
5. Other factors selected at the time of the project
- ii. Gathering the values for the attributes via a short workshop with select management and other personnel
- f. Collect the results and frame a target plan for the balance of the year and into the start of the next year
 - i. Ranking of projects based on attribute criteria
 - ii. Review with key management or department
 - iii. Select the starter projects
- g. Conduct work sessions as needed to do the first implementations
 - i. The team would then develop the short-term AI *road map for pilot projects in specific fields e.g. labor market, HR, IT Operations etc.*
 1. The roadmap would include the target forward planning approach for the following year
 2. The time frame could change, and the forward plan put in place or the short-term plan extended based on intent, support, assets and need.

Deliverables:

1. Stage 1 – Short Term Plan
 - a. List of AI projects
 - b. Assessment of short term opportunities
 - i. Risk
 - ii. Yield
 - iii. Implementation plan of 2 to 3 small projects
 - c. Draft outline for long term forward plan project
2. Stage 2 – Implementation of selected projects starting with one basic project and then expanding as skills and tool use develops
3. Stage 3 – Long term forward plan
 - a. 5 to 20-year plan as determined by AI Ministry direction
 - b. Content from Stage 1 integrated with AI Ministry direction and MOHRE specific needs

Schedule and Costs

Schedule

- Stage1: Start 2 weeks after contract approval. Target output is to get a defined set of small projects that add value to MOHRE
- Stage 2: Start just before stage 1 is complete target to deliver at least 1 solution before the end of the year.
- Stage 3: Start on alignment with annual planning cycle or date determined by MOHRE management